

ACTU 301

Life Contingencies 1

Survival Models

Lecture 2

Age-At-Death Random Variable

The central difficulty in issuing life insurance is that of determining the time of death of the insured.

Let time 0 denote the time of birth of an individual. We will always assume that everyone is alive at birth.

Let X be a continuous and non-negative random variable, then X is an appropriate random variable to model the age at which an individual dies and is therefore called the age at death random variable.

Definition

The cumulative distribution function of X is the probability that a newborn dies before age x and is given by:

$$F_0(x) = \Pr(X \leq x) = \Pr(X < x)$$

That is, $F_0(x)$ is the probability that death will occur prior to (or at) age x .

The distribution function satisfies the following three properties :

- $F_0(0) = 0$,
- $\lim_{x \rightarrow \infty} F_0(x) = 1$,
- $F_0(x)$ is a non-decreasing function.

The Survival Distribution Function of X

In many life insurance problems, one is interested in the probability of survival rather than death, and so we define the survival distribution function by:

$$S_0(x) = \Pr(X > x), \quad x \geq 0$$

This function is also referred to as the survival function

For a function to be a legitimate survival function it must satisfy all three of the following properties.

- $S_0(0) = 1$,
- $\lim_{x \rightarrow \infty} S_0(x) = 0$,
- $S_0(x)$ is a non-increasing function of x .

Since surviving beyond age x and dying before age x are complementary events, there exists a simple relationship between the survival and distribution functions, namely,

$$S_0(x) = 1 - F_0(x)$$

Definition:

Assume that for the r.v. X there exists a continuously differentiable survival function $S_0(x)$, then X has a continuous distribution function $F_0(x)$ and the probability density function may be defined as

$$f_0(x) = \frac{d}{dx} F_0(x)$$

Given the p.d.f there also exist integral expressions for the survival and distribution functions

$$S_0(x) = \int_x^{\infty} f_0(u) du \quad F_0(x) = \int_0^x f_0(u) du$$

using $S_0(x) = 1 - F_0(x)$

one can obtain the following expression for the density in terms of the survival function as follows:

$$f_0(x) = \frac{d}{dx} F_0(x) = \frac{d}{dx} (1 - S_0(x)) = -\frac{d}{dx} S_0(x)$$

Using laws of probability, we can use either $F(x)$ or $S(x)$ to answer probability statements, such as .

$$\begin{aligned} \Pr(a < X \leq b) &= F_0(b) - F_0(a) \\ &= [1 - S_0(b)] - [1 - S_0(a)] \\ &= S_0(a) - S_0(b) \end{aligned}$$

Relations between $f_0(x)$, $F_0(x)$ and $S_0(x)$

$$1. f_0(x) = \frac{d}{dx} F_0(x) = \frac{d}{dx} (1 - S_0(x)) = -\frac{d}{dx} S_0(x)$$

$$2. S_0(x) = 1 - F_0(x)$$

$$3. \Pr(a < X \leq b) = F_0(b) - F_0(a) = S_0(a) - S_0(b)$$

Example 1

Consider the proposed survival function

$$S_0(x) = \left[1 - \frac{x}{100}\right]^{\frac{1}{2}} \quad 0 \leq x \leq 100$$

Determine whether this function is a legitimate survival function

Solution

We need to check that $S_0(x)$ satisfies the three conditions.

By substituting for $x=0$ it can be seen that

(i) for $x=0$, $S_0(0) = 1$

(ii) for $x=100$, $S_0(100) = 0$.

(iii) For the function to be non-increasing we consider the first derivative.

$$S'_0(x) = -\frac{1}{200} \left[1 - \frac{x}{100}\right]^{-\frac{1}{2}}$$

Since $S'_0(x)$ is non-positive for all x in the given domain, $S_0(x)$ must be non-increasing.

All the three properties are satisfied so we conclude that $S_0(x)$ is a legitimate survival function.

Example 2

Consider the survival function

$$S_0(x) = \frac{1}{10}[100 - x]^{\frac{1}{2}} \quad 0 \leq x \leq 100$$

What is the probability that a newborn dies between the ages of 65 and 75?

Solution

The required probability is

$$\begin{aligned} \Pr(65 < X \leq 75) &= F_0(75) - F_0(65) \\ &= [1 - S_0(75)] - [1 - S_0(65)] \\ &= S_0(65) - S_0(75) \end{aligned}$$

$$S_0(65) = \frac{1}{10}[100 - 65]^{\frac{1}{2}} = \frac{1}{10}[35]^{\frac{1}{2}} = 0.591608$$

$$S_0(75) = \frac{1}{10}[100 - 75]^{\frac{1}{2}} = \frac{1}{10}[25]^{\frac{1}{2}} = 0.5$$

$$\begin{aligned} \Pr(65 < X \leq 75) &= S_0(65) - S_0(75) \\ &= 0.591608 - 0.5 = 0.0916 \end{aligned}$$

Force of Mortality of X

The probability density function at x can be used to estimate the probability that death occurs in the interval $(x, x + \Delta x]$ for small Δx given only that the entity existed at age $x = 0$.

That is, the force of mortality considers the probability of death within an infinitesimal interval that we shall denote by Δt

Definition: For a life aged x the force of mortality is defined as

$$\mu_x = \lim_{\Delta x \rightarrow 0} \frac{1}{\Delta x} \Pr(X \leq x + \Delta x | X > x)$$

Let μ_x be a non-negative real-valued function, μ_x may be considered as a **force of mortality** if and only if the following properties are satisfied.

For all $x \geq 0$, $\mu_x \geq 0$

$$\int_0^{\infty} \mu_x dx = \infty$$

From the definition of the force of mortality, we can obtain an equation linking it with both the survival function and p.d.f.

$$\begin{aligned} \mu_x &= \lim_{\Delta x \rightarrow 0} \frac{1}{\Delta x} \Pr(X \leq x + \Delta x | X > x) \\ &= \lim_{\Delta x \rightarrow 0} \frac{1}{\Delta x} \frac{\Pr(X > x) - \Pr(X > x + \Delta x)}{\Pr(X > x)} \\ &= \lim_{\Delta x \rightarrow 0} \frac{1}{\Delta x} \frac{S_0(x) - S_0(x + \Delta x)}{S_0(x)} \\ &= -\frac{1}{S_0(x)} \lim_{\Delta x \rightarrow 0} \frac{S_0(x + \Delta x) - S_0(x)}{\Delta x} \\ &= -\frac{1}{S_0(x)} \frac{d}{dx} S_0(x) \end{aligned}$$

$$\mu_x = \frac{f_0(x)}{S_0(x)} \quad \text{using} \quad f_0(x) = -\frac{d}{dx} S_0(x)$$

Recall that $\frac{d \ln x}{dx} = \frac{1}{x}$ and that by the chain rule

$$\frac{d \ln g(x)}{dx} = \frac{g'(x)}{g(x)} \quad \text{for a real-valued function } g$$

$$\mu_x = \frac{f_0(x)}{S_0(x)} = -\frac{1}{S_0(x)} \frac{d}{dx} S_0(x) = -\frac{d}{dx} \ln S_0(x)$$

or

$$\mu_x = -\frac{S_0'(x)}{S_0(x)} = -\frac{d}{dx} \ln S_0(x)$$

Integrating the equation

$$\mu_x = -\frac{1}{S_0(x)} \frac{d}{dx} S_0(x) = -\frac{d}{dx} \ln S_0(x)$$

between 0 and y gives

$$\int_0^y \mu_x dx = -[\ln S_0(y) - \ln S_0(0)]$$

but since $S_0(0) = 1$, it follows that $\ln S_0(0) = 0$ and therefore

$$\int_0^y \mu_x dx = -\ln S_0(y)$$

Then taking exponentials of both sides yields

$$S_0(y) = \exp\left(-\int_0^y \mu_x dx\right)$$

A simple exchange of variables gives

$$S_0(x) = \exp\left(-\int_0^x \mu_s ds\right)$$

Relations between μ_x , $f_0(x)$, and $S_0(x)$

$$1. \quad \mu_x = \frac{f_0(x)}{S_0(x)}$$

$$2. \quad \mu_x = -\frac{S_0'(x)}{S_0(x)} = -\frac{d}{dx} \ln S_0(x)$$

$$3. \quad S_0(x) = \exp\left(-\int_0^x \mu_s ds\right)$$

Example 3:

The distribution function for a survival model is

$$F_0(x) = 1 - \left(1 - \frac{x}{120}\right)^{\frac{1}{6}}$$

Find an expression for the force of mortality μ_x .

Solution:

The respective survival and density functions are

$$S_0(x) = \left(1 - \frac{x}{120}\right)^{\frac{1}{6}} \quad \text{and} \quad f_0(x) = \frac{1}{720} \left(1 - \frac{x}{120}\right)^{-\frac{5}{6}}$$

$$\text{using} \quad \mu_x = \frac{f_0(x)}{S_0(x)}$$

Therefore, the force of mortality may be written as

$$\mu_x = \frac{1}{720} \left(1 - \frac{x}{120}\right)^{-1} = \frac{1}{6(120 - x)} = \frac{1}{720 - 6x}$$

The future lifetime random variable

We now introduce the random variable representing the time-until-death past a certain age where the person is known to be alive.

First, we let (x) to indicate that a newborn is known to be alive at age x or simply a life aged x . (i.e. $x \geq 0$)

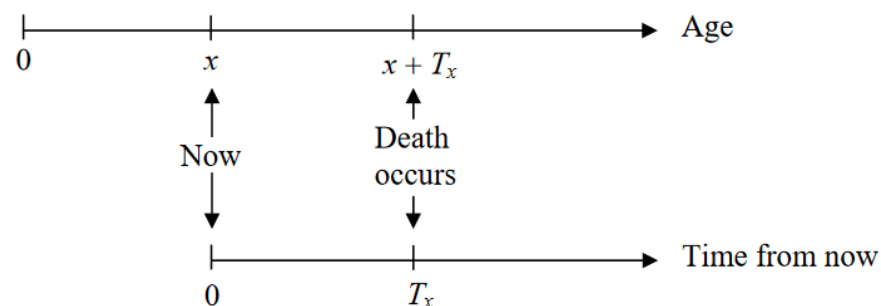
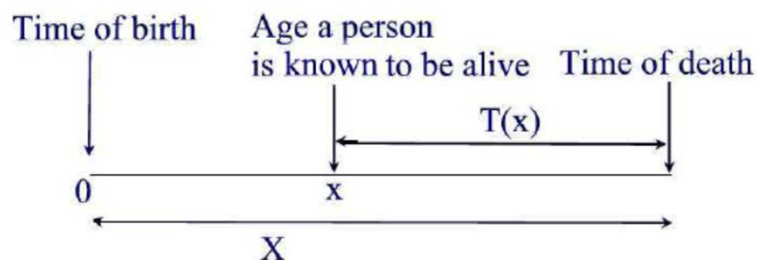
We let $T(x)$ or T_x be the continuous random variable that represents the additional time (x) might survive beyond the age of x .

We refer to this variable as the time-until-death or the future-lifetime random variable.

From this definition, we have a relationship between the age-at-death random variable X and the time-until-death $T(x)$ given by

$$X = x + T(x).$$

A pictorial representation of X , x and $T(x)$ is shown



- ❖ If there is no limiting age, T_x can take any value within $[0, \infty)$.
- ❖ If a limiting age is assumed, then T_x can only take a value within $[0, \omega - x]$.
- ❖ We have to subtract x because the individual has attained age x at time 0 already.

Thus, for someone age 50, T_{50} is the amount of time until he dies, and to say $T_{50} = 32.4$ means that the person who was originally age 50 died when he was age 82.4, so that he lived exactly 32.4 years.

We will use the symbol (x) to mean “someone age x ”, so (50) means “someone age 50”..

Definition:

If X is the age at death random variable and x is the current age of an individual, the future lifetime can be defined by the random variable

$$T_x = X - x$$

Consider an individual currently aged x who survives for another t years, the probability of this is given by the **survival function**

$$S_x(t) = \Pr(T_x > t)$$

The subscript “ x ” indicates that the survival function is defined for a life who is age x now. It is important to understand that when modeling the future lifetime of (x) , we always know that the individual is alive at age x .

Similarly, the probability that the individual dies within t years is given by the **distribution function**

$$F_x(t) = \Pr(T_x \leq t)$$

When we consider $S_x(t)$ we make the underlying assumption that the life has survived from birth through to age x before surviving from age x to $x+t$. This assumption allows us to write $S_x(t)$ in terms of the age at death random variable X and results in a very important equation relating $S_0(x)$ and $S_x(t)$.

Thus $S_{50}(30)$ is the probability that (50) lives at least 30 years.

$F_{50}(30)$ is the probability that (50) does not survive 30 years

$$\begin{aligned} S_x(t) &= \Pr(T_x > t) = \Pr(T_0 > x+t \mid T_0 > x) \\ &= \frac{\Pr(T_0 > x+t \cap T_0 > x)}{\Pr(T_0 > x)} = \frac{\Pr(T_0 > x+t)}{\Pr(T_0 > x)} = \frac{S_0(x+t)}{S_0(x)} \end{aligned}$$

Thus the **Survival Function** for the Future Lifetime Random Variable

$$S_x(t) = \frac{S_0(x+t)}{S_0(x)}$$

So that $S_0(x+t) = S_0(x) S_x(t)$

This is a very important result. It shows that we can interpret the probability of survival from age x to age $x+t$ as the product of:

- ❖ the probability of survival to age x from birth, and
- ❖ the probability, having survived to age x , of further surviving to age $x+t$.

Similarly, any survival probability for (x) , for, say, $t+u$ years can be split into the probability of surviving the first t years, and then, given survival to age $x+t$, subsequently surviving another u years. That is,

$$\begin{aligned} S_x(t+u) &= \frac{S_0(x+t+u)}{S_0(x)} \\ \Rightarrow S_x(t+u) &= \frac{S_0(x+t)}{S_0(x)} \frac{S_0(x+t+u)}{S_0(x+t)} \\ \Rightarrow S_x(t+u) &= S_x(t) S_{x+t}(u). \end{aligned}$$

Survival function conditions and assumptions

We will require any survival function corresponding to a future lifetime random variable to satisfy the following conditions:

Necessary and Sufficient Conditions

Condition 1. $S_x(0) = 1$;

That is, the probability that a life currently aged x survives 0 years is 1.

Condition 2. $\lim_{t \rightarrow \infty} S_x(t) = 0$;

That is, all lives eventually die.

Condition 3. The survival function must be a non-increasing function of t ; it cannot be more likely that (x) survives, say 10.5 years than 10 years, because in order to survive 10.5 years, (x) must first survive 10 years.

These conditions are both necessary and sufficient, so that any function S_x which satisfies these three conditions as a function of t (≥ 0), for a fixed x (≥ 0), defines a lifetime distribution from age x , and, using formula above for all ages greater than x .

We make three additional assumptions

Assumption 1. $S_x(t)$ is differentiable for all $t > 0$.

Assumption 2. $\lim_{t \rightarrow \infty} t S_x(t) = 0$

Assumption 3. $\lim_{t \rightarrow \infty} t^2 S_x(t) = 0$

NOTE

1. The Survival Function for the Future Lifetime Random Variable

$$S_x(t) = \frac{S_0(x+t)}{S_0(x)}$$

2. Any survival probability for (x) , for, say, $t+u$ years can be split into the probability of surviving the first t years, and then, given survival to age $x+t$, subsequently surviving another u years

$$S_x(t+u) = S_x(t) S_{x+t}(u)$$

Actuarial notation

The actuarial notation used to define the survival and distribution functions is as follows

Notation 1: ${}_t p_x$

We use ${}_t p_x$ to denote the probability that a life age x now survives to t years from now. By definition, we have

$${}_t p_x = \Pr[T_x > t] = S_x(t)$$

When $t = 1$, we can omit the subscript on the left-hand-side; that is, we write ${}_1 p_x$ as p_x .

Notation 2: ${}_t q_x$

We use ${}_t q_x$ to denote the probability that a life age x now dies before attaining age $x+t$. By definition, we have

$${}_t q_x = \Pr[T_x \leq t] = 1 - S_x(t)$$

When $t = 1$, we can omit the subscript on the left-hand-side; that is, we write ${}_1 q_x$ as q_x .

Notation 3: ${}_t|u q_x$

We use ${}_t|u q_x$ to denote the probability that a life age x now dies between ages $x+t$ and $x+t+u$. By definition, we have

$${}_t|u q_x = \Pr[t < T_x \leq t+u] = F_x(t+u) - F_x(t) = S_x(t) - S_x(t+u)$$

When $u = 1$, we can omit the subscript u ; that is, we write ${}_t|_1 q_x$ as ${}_t q_x$

- ❖ Note that when we describe survival distributions, “ p ” always means a survival probability, while “ q ” always means a death probability.
- ❖ The “ $|$ ” between t and u means that the death probability is deferred by t years. We read “ $t | u$ ” as “ t deferred u ”.

This is called a **deferred mortality probability**, because it is the probability that death occurs in some interval following a deferred period.

In actuarial terminology q_x is called the **mortality rate** at age x .

Note that ${}_t q_x$ is the probability of (x) does not survive beyond age $x+t$ or the conditional probability that death occurs not later than age $x+t$, given survival to age x .

Some Basic Relationships

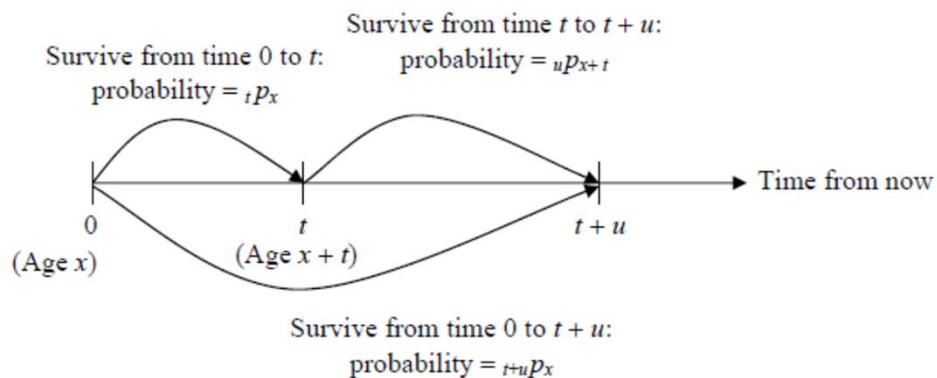
We can express some relationships in this actuarial notation as follows:

Equation 1: ${}_t p_x + {}_t q_x = 1$

This equation arises from the fact that there are only two possible outcomes: dying within t years or surviving to t years from now.

Equation 2: ${}_{t+u} p_x = {}_t p_x \times {}_u p_{x+t}$

The meaning of this equation can be seen from the following diagram.



Mathematically, we can prove this equation as follows:

$$\begin{aligned} {}_{t+u} p_x &= S_x(t+u) = \frac{S_0(x+t+u)}{S_0(x)} = \frac{S_0(x+t)}{S_0(x)} \frac{S_0(x+t+u)}{S_0(x+t)} \\ &= S_x(t) S_{x+t}(u) = {}_t p_x \times {}_u p_{x+t} \end{aligned}$$

Equation 3: ${}_t q_x = {}_{t+u} q_x - {}_t q_x = {}_t p_x - {}_{t+u} p_x$

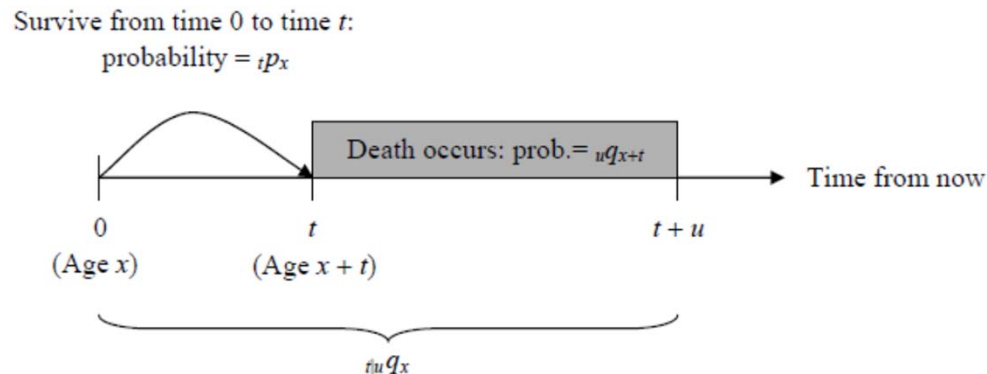
This equation arises naturally from the definition of ${}_t q_x$.

We have ${}_t q_x = \Pr(t < T_x \leq t+u) = F_x(t+u) - F_x(t) = {}_t q_x = {}_{t+u} q_x - {}_t q_x$

Also, ${}_t q_x = \Pr(t < T_x \leq t+u) = S_x(t) - S_x(t+u) = {}_t p_x - {}_{t+u} p_x$.

Equation 4: ${}_t q_x = {}_t p_x \times {}_u q_{x+t}$

The reasoning behind this equation can be understood from the following diagram.



Mathematically, we can prove this equation as follows

$$\begin{aligned}
 {}_t|u q_x &= {}_t p_x - {}_{t+u} p_x \quad (\text{from Equation 3}) \\
 &= {}_t p_x - {}_t p_x \times {}_u p_{x+t} \quad (\text{from Equation 2}) \\
 &= {}_t p_x (1 - {}_u p_{x+t}) \\
 &= {}_t p_x \times {}_u q_{x+t} \quad (\text{from Equation 1})
 \end{aligned}$$

Prove that for any $t \geq 1$,

$${}_t p_x = p_x \cdot {}_{t-1} p_{x+1}.$$

$$\begin{aligned}
 {}_t p_x &= P(X > x+t | X > x) = \frac{P(X > x+t)}{P(X > x)} \\
 &= \frac{P(X > x+1)}{P(X > x)} \cdot \frac{P(X > x+t)}{P(X > x+1)} = P(X > x+1 | X > x) P(X > x+t | X > x+1) \\
 &= P(X > x+1 | X > x) P(X > x+1+t-1 | X > x+1) = p_x \cdot {}_{t-1} p_{x+1}.
 \end{aligned}$$

NOTE Summary of the Relationships

$${}_t p_x = \Pr[T_x > t] = S_x(t)$$

$${}_t q_x = \Pr[T_x \leq t] = 1 - S_x(t)$$

$${}_t|u q_x = \Pr[t < T_x \leq t+u] = S_x(t) - S_x(t+u)$$

$${}_t p_x + {}_t q_x = 1$$

$${}_{t+u} p_x = {}_t p_x \times {}_u p_{x+t}$$

$${}_t|u q_x = {}_{t+u} q_x - {}_t q_x = {}_t p_x - {}_{t+u} p_x$$

$${}_t|u q_x = {}_t p_x \times {}_u q_{x+t}$$

$${}_t p_x = p_x \times {}_{t-1} p_{x+1}$$

The Cumulative Distribution Function of T_x

$$F_x(t) = \Pr[T_x \leq t] = 1 - S_x(t) = {}_tq_x = 1 - {}_tp_x = 1 - \frac{S_x(x+t)}{S_x(x)}$$

$$= \frac{S_x(x) - S_x(x+t)}{S_x(x)} = \frac{F_x(x+t) - F_x(x)}{1 - F_x(x)}$$

Thus the conditional probability that a newborn will die between the ages a and b , given survival to age x , is

$$\Pr(a < X \leq b | X > a) = \frac{\Pr(a < X \leq b)}{\Pr(X > a)}$$

$$= \frac{S_0(a) - S_0(b)}{S_0(a)}$$

Probability density function of T_x

The p.d.f of the future lifetime random variable is given in terms of the distribution and survival functions in a similar way as before.

$$f_x(t) = \frac{d}{dt} F_x(t) = \frac{d}{dt} (1 - S_x(t)) = -\frac{d}{dt} S_x(t)$$

or $f_x(t) = \frac{d}{dt} [{}_tq_x] = \frac{d}{dt} \left[1 - \frac{S_x(x+t)}{S_x(x)} \right] = -\frac{d}{dt} \left(\frac{S_x(x+t)}{S_x(x)} \right)$

$$= -\frac{1}{S_x(x)} \frac{d}{dt} (S_x(x+t)) = \frac{f_x(x+t)}{S_x(x)}$$

In words, $f_x(t)$ is the conditional density of death at time t given survival to age x or the conditional density at age $x+t$, given survival to age x .

Relations between $f_x(t)$, $S_x(t)$ and $F_x(t)$

$$1. f_x(t) = \frac{d}{dt} F_x(t) = -\frac{d}{dt} S_x(t)$$

$$2. f_x(t) = \frac{f_x(x+t)}{S_x(x)}$$

Example 4

An age-at-death random variable has the survival function $s(x) = 1 - \frac{x^2}{10000}$ for $0 \leq x < 100$.

(a) Find the survival function of $T(x)$.

(b) Find the PDF of $T(x)$.

Solution

(a) The survival function of $T(x)$ is

$${}_tp_x = \frac{s(x+t)}{s(x)} = \frac{1 - \frac{(x+t)^2}{10000}}{1 - \frac{x^2}{10000}} = \frac{10000 - (x+t)^2}{10000 - x^2}, \quad 0 \leq t \leq 100 - x$$

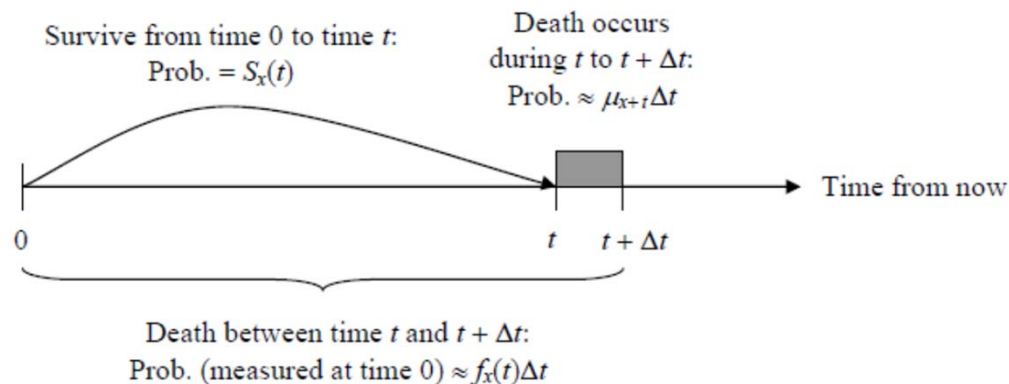
(b) The PDF of $T(x)$ is

$$f_{T(x)}(t) = -\frac{d}{dt} \left(\frac{s(x+t)}{s(x)} \right) = \frac{2(x+t)}{10000 - x^2}, \quad 0 \leq t \leq 100 - x$$

Force of Mortality of T_x

Consider an individual age x now.

- ❖ The force of mortality for this individual t years from now is denoted by μ_{x+t} or $\mu_x(t)$.
- ❖ At time t , the (approximate) probability that this individual dies within a very small period of time Δt is $\mu_{x+t} \Delta t$.
- ❖ The definition of μ_{x+t} can be seen from the following diagram.



From the diagram, we can also tell that $f_x(t) \Delta t = S_x(t) \mu_{x+t} \Delta t$. It follows that

$$f_x(t) = S_x(t) \mu_{x+t} = {}_t p_x \mu_{x+t}$$

Recall that $f_x(t) = F'_x(t) = -S'_x(t)$ or $f_x(t) = \frac{d}{dt} F_x(t) = -\frac{d}{dt} S_x(t)$

This yields the following equation $\mu_{x+t} = \frac{f_x(t)}{S_x(t)} = -\frac{S'_x(t)}{S_x(t)}$

This equation allows us to find the force of mortality when the survival function is known.

The force of mortality may also be defined for the future lifetime random variable as:

$$\begin{aligned} \mu_x(t) &= \lim_{\Delta t \rightarrow 0} \frac{\Pr(t < T_x \leq t + \Delta t | T_x > t)}{\Delta t} \\ &= \lim_{\Delta t \rightarrow 0} \frac{\Pr(x + t < X \leq x + t + \Delta t | X > x + t)}{\Delta t} = \mu(x+t) \end{aligned}$$

That is, $\mu(x+t)$ is the death rate at age $x + t$ given survival to age $x + t$.

Probability density function of T_x using force of Mortality

Let $F_x(t)$ and $f_x(t)$ denote, respectively, the d.f. and p.d.f. of T_x

$$\begin{aligned} f_x(t) &= \frac{d}{dt} F_x(t) = \frac{d}{dt} {}_t q_x \\ &= \frac{d}{dt} \left[1 - \frac{S_0(x+t)}{S_0(x)} \right] \\ &= \frac{S_0(x+t)}{S_0(x)} \left[-\frac{S'_0(x+t)}{S_0(x+t)} \right] = {}_t p_x \mu_{x+t} \end{aligned}$$

That is, the density of the random variable T_x can be written as

$$f_x(t) = {}_t p_x \mu_{x+t} \quad x \geq 0, t \geq 0.$$

Thus ${}_t p_x \mu_{x+t} dt$ is the probability that (x) dies between t and $t + dt$, and

$$\int_0^\infty {}_t p_x \mu_{x+t} dt = 1$$

Also
$$\mu_{x+t} = \frac{f_x(t)}{{}_t p_x} = \frac{f_x(t)}{S_x(t)}$$

$$\mu_{x+t} = \frac{F'_x(t)}{1 - F_x(t)} = \frac{f_x(t)}{S_x(t)}$$

Recall that $\frac{d \ln x}{dx} = \frac{1}{x}$ and that by the chain rule

$$\frac{d \ln g(x)}{dx} = \frac{g'(x)}{g(x)} \quad \text{for a real-valued function } g$$

$$\mu_{x+t} = -\frac{S'_x(t)}{S_x(t)} = -\frac{d}{dt} \ln S_x(t)$$

$$-\mu_{x+t} dt = d[\ln S_x(t)]$$

Replacing t by u , $-\mu_{x+u} du = d[\ln S_x(u)]$

Integrating on both sides yields

$$-\int_0^t \mu_{x+u} du = \int_0^t d[\ln S_x(u)]$$

$$-\int_0^t \mu_{x+u} du = \ln S_x(t) - \ln S_x(0)$$

but since $S_0(0) = 1$, it follows that $\ln S_0(0) = 0$ and therefore

$$-\int_0^t \mu_{x+u} du = \ln S_x(t)$$

Then taking exponentials of both sides yields

$$S_x(t) = \exp\left(-\int_0^t \mu_{x+u} du\right)$$

This allows us to find the survival function when the force of mortality is known.

Note: $S_x(t) = \frac{S_0(x+t)}{S_0(x)} = \exp\left(-\int_0^t \mu_{x+u} du\right)$

The force of mortality must satisfy the following two criteria:

(i) $\mu_{x+t} \geq 0$ for all $x \geq 0$ and $t \geq 0$.

(ii) $\int_0^\infty \mu_{x+u} du = \infty$.

Alternatively

Let $\mu_x(t)$ be the force of mortality for the r.v. T_x , and let $f_x(t)$ be the density of T_x . Also let $\mu(x)$ be the force of mortality for X . Then

$$(i) \mu_x(t) = \mu(x+t) = \mu_{x+t}$$

$$(ii) f_x(t) = \mu_x(t) {}_t p_x = \mu(x+t) {}_t p_x = \mu_{x+t} {}_t p_x$$

since ${}_t q_x$ is the d.f. of T_x , the density

$$f_x(t) = \frac{d}{dt} {}_t q_x = \frac{d}{dt} (1 - {}_t p_x) = -\frac{d}{dt} {}_t p_x = \mu(x+t) \exp \left\{ -\int_x^{x+t} \mu(z) dz \right\}$$

Since the exponent above equals ${}_t p_x$, we have $f_x(t) = \mu(x+t) {}_t p_x$. On the other hand, by the definition of force of mortality,

$$\mu_x(t) = \frac{f_x(t)}{P(T_x > t)} = \frac{f_x(t)}{{}_t p_x} = \frac{\mu(x+t) {}_t p_x}{{}_t p_x} = \mu(x+t) = \mu_{x+t}$$

Other frequently used probability characteristics

$q_x = {}_1 q_x = P(T_x \leq 1)$, the probability that a life-age- x will die within one year;

$p_x = {}_1 p_x = P(T_x > 1)$, the probability that a life-age- x will live at least one year more;

${}_t u q_x = P(t < T_x \leq t+u) = P(x+t < X \leq x+t+u \mid X > x)$.

Relations between μ_{x+t} , $f_x(t)$, $S_x(t)$,

$$1. f_x(t) = S_x(t) \mu_{x+t} = {}_t p_x \mu_{x+t}$$

$$2. \mu_{x+t} = \frac{f_x(t)}{{}_t p_x} = \frac{f_x(t)}{S_x(t)} = -\frac{S'_x(t)}{S_x(t)}$$

$$3. S_x(t) = {}_t p_x = \exp \left(-\int_0^t \mu_{x+u} du \right)$$

The following two specifications of the force of mortality are often used in practice

Gompertz' law

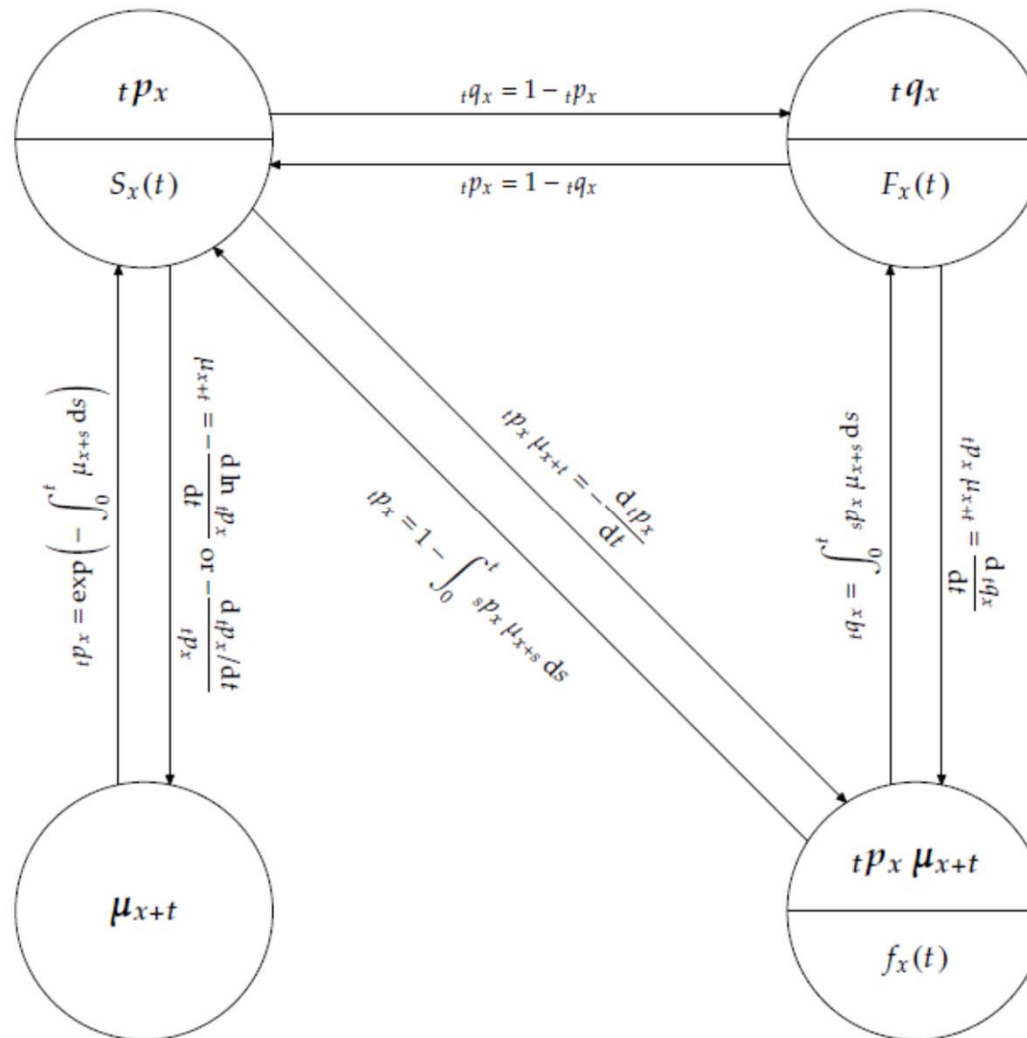
$$\mu_x = Bc^x$$

Makeham's law

$$\mu_x = A + Bc^x$$

A , B and c are constants such that $A \geq -B$, $B > 0$ and $c > 1$.

Relationships between ${}_t p_x$, ${}_t q_x$, and μ_{x+t}



Expected future lifetime

Like any random variable, the expected value of T_x , known as the complete expectation of life or expected future lifetime and denoted in actuarial notation by e_x° is defined as:

$$e_x^\circ = E[T_x] = \int_0^\infty t f_x(t) dt$$

We can also write
$$e_x^\circ = \int_0^\infty t {}_t p_x \mu(x+t) dt$$

From the definition of an expected value, we have

$$\begin{aligned} E[T_x] &= e_x^\circ = \int_0^\infty t f_x(t) dt \\ &= \int_0^\infty t {}_t p_x \mu_{x+t} dt \\ &= \int_0^\infty {}_t p_x dt \end{aligned}$$

That is, e_x° is called the complete expectation of life at age x:

$$\begin{aligned} e_x^\circ &= E[T_x] = \int_0^\infty t f_x(t) dt \\ &= \int_0^\infty t \left(-\frac{d}{dt} {}_t p_x \right) dt \\ &= [t \cdot {}_t p_x]_0^\infty - \int_0^\infty (-{}_t p_x) dt. \end{aligned}$$

Since ${}_t p_x \rightarrow 0$ as $t \rightarrow \infty$, evaluating the first term at each end point yields zero and so the only non-zero term is the second term.

Therefore,

$$e_x^\circ = \int_0^\infty {}_t p_x dt$$

Alternatively in terms of survival function,

$$\begin{aligned} e_x^\circ &= \int_0^\infty t f(t) dt \\ &= \int_0^\infty t \frac{-S'(x+t)}{S(x)} dt \\ &= -\frac{1}{S(x)} \int_0^\infty t S'(x+t) dt \\ &= -\frac{1}{S(x)} \left\{ [tS(x+t)]_{t=0}^{t=\infty} - \int_0^\infty S(x+t) dt \right\} \quad [\text{integration by parts}] \\ &= \int_0^\infty \frac{S(x+t)}{S(x)} dt \quad [\text{assuming } xS(x) \rightarrow 0 \text{ as } x \rightarrow \infty] \\ &= \int_0^\infty {}_t p_x dt \end{aligned}$$

Next, the variance of T_x is given by

$$Var(T_x) = E[T_x^2] - [E(T_x)]^2$$

$$\text{where } E(T_x) = \int_0^{\infty} \frac{S(x+t)}{S(x)} dt = \int_0^{\infty} {}_t p_x dt$$

The second moment can be evaluated by considering

$$E[T_x^2] = \int_0^{\infty} t^2 f_x(t) dt$$

which may be again evaluated using integration by parts to give

$$\begin{aligned} E(T_x^2) &= \int_0^{\infty} t^2 f_x(t) dt = \int_0^{\infty} t^2 \frac{f_x(x+t)}{S(x)} dt \\ &= \frac{1}{S(x)} \int_0^{\infty} t^2 f_x(x+t) dt \\ &= -\frac{1}{S(x)} \left\{ [-t^2 S(x+t)]_{t=0}^{t=\infty} + \int_0^{\infty} 2t S(x+t) dt \right\} \\ &= \int_0^{\infty} \frac{2t S(x+t)}{S(x)} dt \\ &= \int_0^{\infty} 2t {}_t p_x dt \end{aligned}$$

Example 5

Let the age-at-death X be exponential with density function

$$f(x) = 0.05e^{-0.05x}, \quad x \geq 0$$

Calculate the variance of T_x .

Solution

$$F(x) = 1 - e^{-0.05x}$$

$$S(x) = e^{-0.05x}$$

$$E(T_x) = \int_0^{\infty} {}_t p_x dt = \int_0^{\infty} e^{-0.05t} dt = 20$$

$$E(T_x^2) = \int_0^{\infty} 2t {}_t p_x dt = \int_0^{\infty} 2te^{-0.05t} dt = 80$$

$$Var(T_x) = E[T_x^2] - [E(T_x)]^2 = 80 - 20^2 = 40$$

Curtate future lifetime random variable

T_x is a continuous-type random variable which expresses the *exact* time-until-death for (x) . One can also associate with the future lifetime of (x) , a discrete-type random variable, the number of further years *completed* by (x) prior to death which is denoted by K_x and known as the **curtate future lifetime**.

The curtate future lifetime r.v is defined as the integer part of the future lifetime r.v and can be thought of as the number of whole years the individual has survived for.

Definition:

For a life aged x , if T_x is the future lifetime random variable, the curtate future lifetime random variable is defined as the integer part of future lifetime and is denoted by K_x for a life aged x (K stands for kurtate). K_x is a discrete random variable; in fact K_x assumes integer values only

$$K_x = \lfloor T_x \rfloor$$

where $\lfloor x \rfloor$ is the greatest integer less than or equal to x .

For example, $\lfloor 1 \rfloor = 1$, $\lfloor 4.3 \rfloor = 4$ and $\lfloor 10.99 \rfloor = 10$. We call K_x the curtate future lifetime random variable.

We can think of the curtate future lifetime as the number of whole years lived in the future by an individual.

The probability mass function (p.m.f) for K_x can be expressed as

$$\Pr(K_x = k) = {}_kq_x$$

$K_x = k$ if and only if (x) dies between the ages of $x + k$ and $x + k + 1$. Thus for $k = 0, 1, 2, \dots$

$$\begin{aligned} \Pr(K_x = k) &= \Pr(k \leq T_x < k + 1) \\ &= \Pr(k < T_x \leq k + 1) \quad [\text{since } T(x) \text{ is continuous}] \\ &= \Pr(T_x > k) - \Pr(T_x > k + 1) \\ &= \frac{S(x + k) - S(x + k + 1)}{S(x)} \\ &= {}_kp_x - {}_{k+1}p_x \\ &= {}_k|1q_x \\ &= {}_k|q_x \\ &= {}_kp_x - {}_kp_x p_{x+k} \\ &= {}_kp_x q_{x+k}. \end{aligned}$$

The distribution function for K_x is given by

$$\Pr(K_x \leq k) = {}_{k+1}q_x$$

$$\begin{aligned} \Pr(K_x \leq K) &= \Pr(K_x = 0) + \Pr(K_x = 1) + \dots + \Pr(K_x = k) \\ &= {}_0q_x + {}_1q_x + \dots + {}_kq_x \\ &= \sum_{j=0}^k {}_jq_x = \sum_{j=0}^k ({}_jp_x - {}_{j+1}p_x) \\ &= \sum_{j=0}^k {}_jp_x - \sum_{j=0}^k {}_{j+1}p_x \\ &= ({}_0p_x + {}_1p_x + \dots + {}_kp_x) - ({}_1p_x + {}_2p_x + \dots + {}_{k+1}p_x) \\ &= 1 - {}_{k+1}p_x \quad (\text{since } {}_0p_x = 1) \\ &= {}_{k+1}q_x \end{aligned}$$

We do not derive an expression for the survival function of K_x as this can easily be deduced from knowing the distribution function.

Curtate expectation of life

The expected value of K_x is denoted by e^x , so that $e^x = E[K_x]$, and is referred to as the curtate expectation of life (even though it represents the expected curtate lifetime). So

$$\begin{aligned} E[K_x] &= \sum_{k=0}^{\infty} k \Pr(K_x = k) = \sum_{k=0}^{\infty} k \cdot {}_kq_x \\ &= \sum_{k=0}^{\infty} k ({}_kp_x - {}_{k+1}p_x) \\ &= \sum_{k=0}^{\infty} k \cdot {}_kp_x - \sum_{k=0}^{\infty} k \cdot {}_{k+1}p_x \\ &= ({}_1p_x + 2{}_2p_x + 3{}_3p_x + \dots) - ({}_2p_x + 2{}_3p_x + 3{}_4p_x + \dots) \\ &= {}_1p_x + {}_2p_x + {}_3p_x + \dots \\ &= \sum_{k=1}^{\infty} {}_kp_x \end{aligned}$$

Therefore

$$e_x = \sum_{k=1}^{\infty} {}_kp_x$$

If there is a limiting age, we replace ∞ with $\omega-x$

Alternatively

$$e_x = E(K_x) = \sum_{k=0}^{\infty} k \Pr(K_x = k)$$

We can express e_x in terms of the survival function as follows:

$$\begin{aligned} e_x &= E(K_x) = \sum_{k=0}^{\infty} k \Pr(K_x = k) \\ &= \sum_{k=0}^{\infty} k \left[\frac{s(x+k) - s(x+k+1)}{s(x)} \right] \\ &= \frac{1}{s(x)} \left[\sum_{k=0}^{\infty} k s(x+k) - \sum_{k=0}^{\infty} k s(x+k+1) \right] \\ &= \frac{1}{s(x)} \left[\left(0 + \sum_{k=1}^{\infty} k s(x+k) \right) - \sum_{j=1}^{\infty} (j-1) s(x+j) \right] \quad [\text{using } j = k+1] \\ &= \frac{1}{s(x)} \left[\sum_{k=1}^{\infty} k s(x+k) - \sum_{k=1}^{\infty} (k-1) s(x+k) \right] \quad [\text{relabelling } j \text{ as } k] \\ &= \frac{1}{s(x)} \sum_{k=1}^{\infty} [k - (k-1)] s(x+k) \\ &= \frac{1}{s(x)} \sum_{k=1}^{\infty} s(x+k). \end{aligned}$$

Therefore

$$e_x = \sum_{k=1}^{\infty} {}_k p_x$$

$$E(K_x^2) = \sum_{k=0}^{\infty} k^2 \Pr(K_x = k)$$

$$\begin{aligned} E[K_x^2] &= \sum_{k=0}^{\infty} k^2 ({}_k p_x - {}_{k+1} p_x) \\ &= (1p_x - 2p_x) + 4(2p_x - 3p_x) + 9(3p_x - 4p_x) + 16(4p_x - 5p_x) + \dots \\ &= 2 \sum_{k=1}^{\infty} k {}_k p_x - \sum_{k=1}^{\infty} k p_x = \sum_{k=1}^{\infty} (2k-1) {}_k p_x \\ &= 2 \sum_{k=1}^{\infty} k {}_k p_x - e_x. \end{aligned}$$

Alternatively

$$\begin{aligned} E(K_x^2) &= \sum_{k=0}^{\infty} k^2 \Pr(K_x = k) = \sum_{k=0}^{\infty} k^2 {}_k q_x \\ &= 0^2 \times q_x + 1^2 \times {}_1 q_x + 2^2 \times {}_2 q_x + 3^2 \times {}_3 q_x + \dots \\ &= (p_x - {}_2 p_x) + 4({}_2 p_x - {}_3 p_x) + 9({}_3 p_x - {}_4 p_x) + \dots \\ &= p_x + 3{}_2 p_x + 5{}_3 p_x + \dots \\ &= \sum_{k=1}^{\infty} (2k-1) {}_k p_x. \end{aligned}$$

if there is a limiting age, we replace ∞ with $\omega - x$.

Given the two formulas above, we can easily obtain $Var(K_x)$.

Recursive formula

e_x and e_{x+1} are related to each other as follows:

$$e_x = p_x(1 + e_{x+1}) \quad \text{Recursive formula}$$

$$\begin{aligned} \text{proof } e_x &= \sum_{k=1}^{\infty} {}_k p_x = p_x + \sum_{k=2}^{\infty} {}_k p_x \\ &= p_x + p_x \sum_{k=2}^{\infty} {}_{k-1} p_{x+1} = p_x + p_x \sum_{j=1}^{\infty} {}_j p_{x+1} \\ &= p_x + p_x e_{x+1} = p_x(1 + e_{x+1}) \end{aligned}$$

Relation between complete and curtate expectations of life

Curtate life expectancy ignores the fraction of the last year of life, while complete life expectancy includes it.

That fraction of the last year can range from 0 to 1, but it tends to be about 1/2 for a reasonable mortality function.

Therefore, curtate future lifetime expectancy is about one half year less than complete future lifetime expectancy.

If future lifetime is uniformly distributed and the limiting age ω is an integral number of years away, this relationship is exact.

In symbols, the following equation holds for uniform mortality:

$$e_x^{\circ} = e_x + \frac{1}{2} \quad \text{if } \omega - x \text{ is a non-negative integer}$$

Temporary expected future lifetime

The temporary expectation of life is the number of years an individual is expected to survive between the ages of x and $x+n$, $n > 0$.

The temporary expectation of life is denoted by $e_{x:\overline{n}|}^{\circ}$

and the curtate temporary expectation of life by $e_{x:\overline{n}|}$.

The formulae remain the same as before with the exception of the upper limit which changes from infinity to n , the formulae are therefore

$$e_{x:\overline{n}|}^{\circ} = \int_0^n {}_t p_x dt \quad e_{x:\overline{n}|} = \sum_{k=1}^n {}_k p_x$$

curtate temporary expectation of life

$$\begin{aligned} e_{x:\overline{n}|} &= \sum_{k=0}^{n-1} k \Pr(K_x = k) + \sum_{k=n}^{\infty} n \Pr(K_x = k) \\ &= \sum_{k=0}^{n-1} k \times {}_k q_x + \sum_{k=n}^{\infty} n \times {}_k q_x \\ &= 0 + 1 \times {}_1 q_x + 2 \times {}_2 q_x + \dots + (n-1) \times {}_{n-1} q_x + n \times {}_n q_x + n \times {}_{n+1} q_x + \dots \\ &= ({}_0 p_x - {}_1 p_x) + 2({}_1 p_x - {}_2 p_x) + \dots + (n-1)({}_{n-1} p_x - {}_n p_x) + \dots \\ &\quad + n({}_n p_x - {}_{n+1} p_x) + n({}_{n+1} p_x - {}_{n+2} p_x) + n({}_{n+2} p_x - {}_{n+3} p_x) + \dots \\ &= {}_0 p_x + {}_1 p_x + {}_2 p_x + \dots + {}_n p_x \\ &= \sum_{k=1}^n {}_k p_x \end{aligned}$$

under uniform mortality:

$$e_{x:\overline{n}|} = e_{x:\overline{n}|}^{\circ} + 0.5 {}_n q_x \quad \text{if } n \text{ is an integer and } x+n \leq \omega$$

Summary: Curtate expectation of life

$$e_x = \sum_{k=1}^{\infty} {}_k p_x.$$

$$E(K_x^2) = \sum_{k=1}^{\infty} (2k-1) {}_k p_x.$$

$$e_{x:\overline{n}|} = \sum_{k=1}^n {}_k p_x$$

$$e_x = p_x(1 + e_{x+1}).$$

$${}^{\circ}e_x = e_x + \frac{1}{2}$$

Summary: complete expectation of life

$${}^{\circ}e_x = \int_0^{\infty} {}_t p_x dt$$

$$E(T_x^2) = 2 \int_0^{\infty} {}_t p_x dt$$

$${}^{\circ}e_{x:\overline{n}|} = \int_0^n {}_t p_x dt$$

Example 6

You are given the following mortality table:

x	q_x
90	0.10
91	0.12
92	0.15
93	0.20
94	1.00

(a) Calculate the curtate life expectancy of (90).

(b) Calculate the expected future lifetime (90)

Solution

(a) Calculate ${}_t p_{90}$ as cumulative products of $1 - q_x$:

$$p_{90} = 0.90$$

$${}_2 p_{90} = (0.90)(0.88) = 0.792$$

$${}_3 p_{90} = (0.792)(0.85) = 0.6732$$

$${}_4 p_{90} = (0.6732)(0.80) = 0.53856$$

$${}_5 p_{90} = 0$$

$$e_{90} = \sum_{k=90}^{95} {}_k p_x$$

$$= p_{90} + {}_2 p_{90} + {}_3 p_{90} + {}_4 p_{90} + {}_5 p_{90}$$

$$= 0.90 + 0.792 + 0.6732 + 0.53856 + 0 = \mathbf{2.90376}$$

(b)

$$\text{using } {}^{\circ}e_x = e_x + \frac{1}{2}$$

The expected future lifetime (90)

$$e_{90}^0 = \frac{1}{2} + e_{90}$$

$$e_{90}^0 = \frac{1}{2} + 2.90376 = \mathbf{3.40376}$$

Example 7

Given the following excerpt of a life table:

x	95	96	97	98
l_x	400	300	100	0

Calculate the following:

- (a) e_{95}
- (b) $\text{Var}(K_{95})$
- (c) $e_{95:\overline{1}|}$
- (d) ${}^{\circ}e_{95}$, assuming UDD
- (e) e_{96} , using the recursion formula

Solution

$$(a) \quad e_{95} = \sum_{k=1}^3 {}_k p_{95} = \frac{l_{96}}{l_{95}} + \frac{l_{97}}{l_{95}} + \frac{l_{98}}{l_{95}} = \frac{300}{400} + \frac{100}{400} + \frac{0}{400} = 1$$

(b) We have

$$\begin{aligned} E(K_{95}^2) &= \sum_{k=1}^3 (2k-1) {}_k p_{95} \\ &= \frac{l_{96}}{l_{95}} + 3 \times \frac{l_{97}}{l_{95}} + 5 \times \frac{l_{98}}{l_{95}} \\ &= \frac{300}{400} + 3 \times \frac{100}{400} + 5 \times \frac{0}{400} \\ &= 1.5 \end{aligned}$$

$$\text{Hence, } \text{Var}(K_{95}) = 1.5 - 1^2 = 0.5$$

$$(c) \quad e_{95:\overline{1}|} = \sum_{k=1}^1 {}_k p_{95} = \frac{l_{96}}{l_{95}} = \frac{300}{400} = 0.75$$

$$(d) \quad \text{Assuming UDD, } {}^{\circ}e_{95} = e_{95} + 0.5 = 1 + 0.5 = 1.5$$

(e) Using the recursion formula

$$e_{95} = p_{95}(1 + e_{96}) = 0.75(1 + e_{96})$$

Therefore

$$e_{96} = 1/0.75 - 1 = 0.3333$$

Example 8

Suppose that $s(x) = e^{-\mu x}$, $x \geq 0$. Find e_x and ${}^\circ e_x$.

Solution.

We have

$$\begin{aligned} {}_t p_x &= \frac{s(x+t)}{s(x)} = e^{-\mu t} \\ {}^\circ e_x &= \int_0^\infty {}_t p_x dt = \int_0^\infty e^{-\mu t} dt = \frac{1}{\mu} \\ e_x &= \sum_{k=1}^\infty {}_k p_x = \sum_{k=1}^\infty e^{-\mu k} \\ &= \frac{e^{-\mu}}{1 - e^{-\mu}} \blacksquare \end{aligned}$$

Complete Future Life time

$$e_x = \int_0^{\infty} t {}_t p_x \mu_{x+t} dt$$

$$\bar{e}_x = \int_0^{\infty} {}_t p_x dt$$

$$\bar{e}_x = \frac{1}{\mu} \quad \text{for constant force of mortality}$$

$$\bar{e}_x = \frac{\omega - x}{2} \quad \text{for uniform}$$

n -year Temporary Complete Future Lifetime

$$e_{x:\overline{n}|} = \int_0^n t {}_t p_x \mu_{x+t} dt + n {}_n p_x$$

$$\bar{e}_{x:\overline{n}|} = \int_0^n {}_t p_x dt$$

$$\bar{e}_{x:\overline{n}|} = {}_n p_x (n) + {}_n q_x (n/2) \quad \text{for uniform}$$

$$\bar{e}_{x:\overline{n}|} = p_x + 0.5q_x \quad \text{for uniform}$$

Curtate Future Lifetime

$$e_x = \sum_{k=1}^{\infty} k {}_k|q_x$$

$$e_x = \sum_{k=1}^{\infty} k p_x$$

$$e_x = \bar{e}_x - 0.5 \quad \text{for uniform}$$

$$E[(K_x)^2] = \sum_{k=1}^{\infty} (2k-1) {}_k p_x$$

$$\text{Var}(K_x) = \text{Var}(T_x) - \frac{1}{12} \quad \text{for uniform}$$

n -year Temporary Curtate Future Lifetime

$$e_{x:\overline{n}|} = \sum_{k=1}^{n-1} k {}_k|q_x + n {}_n p_x$$

$$e_{x:\overline{n}|} = \sum_{k=1}^n k p_x$$

$$e_{x:\overline{n}|} = \bar{e}_{x:\overline{n}|} - 0.5 {}_n q_x \quad \text{for uniform}$$

END